Notebook 2- Feature Relationship Analysis

Section-1 Objective

This notebook investigates the statistical relationships between neighbourhood-level socioeconomic indicators and low-income prevalence within the City of Toronto.

The objective is to quantify the strength and direction of the relationships between the selected independent variables and the dependent variable, **Low_Income_AfterTax_Pct**, prior to predictive modelling.

The analysis combines descriptive statistics, Pearson and Spearman correlation analysis, graphical exploration, and multicollinearity assessment to identify the socioeconomic indicators most strongly associated with neighbourhood vulnerability.

The findings from this notebook provide the statistical evidence required to answer **Research Question 1** and guide feature selection for the predictive modelling stage.

Section 2. Project Input & Output

Input

Dataset

- Machine learning-ready dataset produced in **Notebook 1**
- File: `neighbourhood_profiles_2016_ml_ready.csv`

The dataset contains standardized socioeconomic indicators for the City of Toronto and its 140 neighbourhoods. Each row represents one neighbourhood, while each column represents either a socioeconomic predictor variable or the target variable, **Low_Income_AfterTax_Pct**.

Output

This notebook will produce:

- Descriptive statistical summary of all selected variables
- Pearson correlation analysis
- Spearman rank correlation analysis
- Correlation heatmap
- Ranked socioeconomic predictors based on their association with low-income prevalence
- Scatterplots of the strongest relationships

- Variance Inflation Factor (VIF) analysis for multicollinearity assessment
- Interpretation of statistically significant findings
- Evidence-based answer to **Research Question 1**

Dataset

- File: 'neighbourhood_profiles_2016_analysis_ready.csv' '02_Statistical_Analysis_Results.xlsx'

In [77]:
```
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

from pathlib import Path

from scipy.stats import pearsonr
from scipy.stats import spearmanr

pd.set_option("display.max_columns", None)
pd.set_option("display.float_format", "{:.2f}".format)
```

In [78]:
```
from pathlib import Path

project_root = Path.cwd().parents[1]

print(project_root)
```

```
c:\Users\ksalyamova\Desktop\CIND 820\CIND820-Capstone-Project
```

Section-3 Load Machine Learning Dataset

In [79]:
```
input_path = (project_root/ "data"/ "processed"/ "neighbourhood_profiles_2016_ml_re

df = pd.read_csv(input_path)

print("=" * 60)
print("DATASET LOADED")
print("=" * 60)

print(f"Dataset loaded from:\n{input_path}")
```

```
============================================================
DATASET LOADED
============================================================
Dataset loaded from:
c:\Users\ksalyamova\Desktop\CIND 820\CIND820-Capstone-Project\data\processed\neighbo
urhood_profiles_2016_ml_ready.csv
```

Section 4- Dataset Validation

Before conducting statistical analyses, the machine learning-ready dataset is inspected to verify its structure, dimensions, variable names, data types, and overall completeness.

This verification step confirms that the exported dataset from Notebook 1 has been loaded correctly and is suitable for subsequent statistical analyses. It also provides an opportunity to identify any unexpected structural issues prior to correlation analysis.

In [80]:
```
#Dataset Dimensions

print("=" * 60)
print("DATASET DIMENSIONS")
print("=" * 60)

print(f"Rows: {df.shape[0]}")
print(f"Columns: {df.shape[1]}")
```

```
============================================================
DATASET DIMENSIONS
============================================================
Rows: 140
Columns: 14
```

In [81]:
```
# Dataset Preview

display(df.head())
```

	Neighbourhood	Population	Participation_Rate	Employment_Rate	Unemployment_Rate	E
0	Agincourt North	29113	55.40	50.00	9.80	
1	Agincourt South-Malvern West	23757	59.00	53.20	9.80	
2	Alderwood	12054	66.50	62.40	6.10	
3	Annex	30526	70.60	65.80	6.70	
4	Banbury-Don Mills	27695	59.90	55.60	7.20	

In [82]:
```
# Data Types
print("=" * 60)
print("DATA TYPES")
print("=" * 60)

display(df.dtypes)
```

```
============================================================
DATA TYPES
============================================================
```

```
Neighbourhood             object
Population                 int64
Participation_Rate       float64
Employment_Rate          float64
Unemployment_Rate        float64
Bachelors_Degree           int64
No_Diploma                 int64
Immigrant_Population       int64
Visible_Minority           int64
Renters                    int64
Core_Housing_Need          int64
Unaffordable_Housing     float64
Lone_Parent_Families       int64
Low_Income_AfterTax_Pct  float64
dtype: object
```

In [83]:
```
# Missing Values

print("=" * 60)
print("MISSING VALUES")
print("=" * 60)

display(df.isna().sum())
```

```
============================================================
MISSING VALUES
============================================================
Neighbourhood              0
Population                 0
Participation_Rate         0
Employment_Rate            0
Unemployment_Rate          0
Bachelors_Degree           0
No_Diploma                 0
Immigrant_Population       0
Visible_Minority           0
Renters                    0
Core_Housing_Need          0
Unaffordable_Housing       0
Lone_Parent_Families       0
Low_Income_AfterTax_Pct    0
dtype: int64
```

In [84]:
```
# Dataset Information

df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 140 entries, 0 to 139
Data columns (total 14 columns):
 #   Column                   Non-Null Count  Dtype  
---  ------                   --------------  -----  
 0   Neighbourhood            140 non-null    object 
 1   Population               140 non-null    int64  
 2   Participation_Rate       140 non-null    float64
 3   Employment_Rate          140 non-null    float64
 4   Unemployment_Rate        140 non-null    float64
 5   Bachelors_Degree         140 non-null    int64  
 6   No_Diploma               140 non-null    int64  
 7   Immigrant_Population     140 non-null    int64  
 8   Visible_Minority         140 non-null    int64  
 9   Renters                  140 non-null    int64  
 10  Core_Housing_Need        140 non-null    int64  
 11  Unaffordable_Housing     140 non-null    float64
 12  Lone_Parent_Families     140 non-null    int64  
 13  Low_Income_AfterTax_Pct  140 non-null    float64
dtypes: float64(5), int64(8), object(1)
memory usage: 15.4+ KB
```

Interpretation

The inspection confirms that the machine learning-ready dataset was successfully loaded and retains the structure established in Notebook 1.

The dataset contains one observation for each of the 140 neighbourhoods, with standardized socioeconomic predictor variables and a single depenent variable representing neighbourhood-level low-income prevalence.

All variables exhibit the expected data types, and no missing values were detected. These results confirm that the dataset is complete and suitable for the statistical analyses presented in the following sections.

Section-7. Population-Based Feature Engineering

Several socioeconomic indicators in the machine learning-ready dataset are recorded as absolute population counts rather than proportions. Directly comparing these count-based variables across neighbourhoods of different population sizes may produce misleading statistical relationships driven primarily by neighbourhood size rather than underlying socioeconomic conditions.

To improve comparability, selected count-based independent variables are transformed into population-based percentages by dividing each count by the corresponding neighbourhood population and multiplying by 100. This feature engineering step standardizes the selected variables while preserving their substantive interpretation.

The resulting percentage-based variables provide a more appropriate basis for subsequent correlation analysis, multicollinearity assessment, dimensionality reduction, clustering, and predictive modelling.

In [85]:
```
count_variables = [
    "Bachelors_Degree",
    "No_Diploma",
    "Immigrant_Population",
    "Visible_Minority",
    "Renters",
    "Lone_Parent_Families"]

for column in count_variables:
    df[column + "_Pct"] = (df[column] / df["Population"]) * 100

print("=" * 60)
print("FEATURE NORMALIZATION COMPLETE")
print("=" * 60)

display(
    df[[
            "Neighbourhood",
            "Population",
            "Bachelors_Degree",
            "Bachelors_Degree_Pct",
            "Immigrant_Population",
            "Immigrant_Population_Pct"
        ]    ].head())
```

```
============================================================
FEATURE NORMALIZATION COMPLETE
============================================================
```

	Neighbourhood	Population	Bachelors_Degree	Bachelors_Degree_Pct	Immigrant_Populatic
0	Agincourt North	29113	4380	15.04	1999
1	Agincourt South-Malvern West	23757	4210	17.72	1509
2	Alderwood	12054	1660	13.77	397
3	Annex	30526	9135	29.93	827
4	Banbury-Don Mills	27695	6500	23.47	1319

In [86]:
```
# Replace count-based variables with percentage-based variables

df = df.drop(columns=[
    "Bachelors_Degree",
    "No_Diploma",
    "Immigrant_Population",
    "Visible_Minority",
```

```
    "Renters",
    "Lone_Parent_Families"])

print(df.columns)
```

```
Index(['Neighbourhood', 'Population', 'Participation_Rate', 'Employment_Rate',
       'Unemployment_Rate', 'Core_Housing_Need', 'Unaffordable_Housing',
       'Low_Income_AfterTax_Pct', 'Bachelors_Degree_Pct', 'No_Diploma_Pct',
       'Immigrant_Population_Pct', 'Visible_Minority_Pct', 'Renters_Pct',
       'Lone_Parent_Families_Pct'],
      dtype='object')
```

Interpretation

The selected count-based independent variables were successfully transformed into population-based percentages, reducing the influence of differences in neighbourhood population size while improving comparability across Toronto neighbourhoods.

This feature engineering step ensures that the subsequent statistical analyses examine socioeconomic characteristics relative to neighbourhood population rather than absolute counts. The resulting percentage-based variables provide a more meaningful basis for correlation analysis, dimensionality reduction, clustering, and predictive modelling conducted in the subsequent notebooks.

Section-8. Descriptive Statistics

Before quantifying the relationships between socioeconomic indicators and neighbourhood-level low-income prevalence, descriptive statistics are examined to summarize the characteristics of each variable.

This preliminary analysis provides measures of central tendency, variability, and distribution, allowing an initial assessment of the socioeconomic conditions represented within the dataset.

The descriptive statistics also provide context for the subsequent Pearson and Spearman correlation analyses by identifying the range and variability of each predictor and the target variable.

In [87]:

```
print("=" * 60)
print("DESCRIPTIVE STATISTICS")
print("=" * 60)

display(df.describe().T.round(2))
```

```
============================================================
DESCRIPTIVE STATISTICS
============================================================
```

	count	mean	std	min	25%	50%	75%
Population	140.00	19511.22	10033.59	6577.00	12019.50	16749.50	23854.50
Participation_Rate	140.00	64.55	6.11	52.20	59.88	63.60	69.12
Employment_Rate	140.00	59.27	6.52	47.30	54.68	58.55	64.30
Unemployment_Rate	140.00	8.30	1.90	4.50	6.90	8.20	9.62
Core_Housing_Need	140.00	1720.14	1002.88	115.00	967.50	1432.50	2287.50
Unaffordable_Housing	140.00	35.07	6.94	17.50	31.08	35.60	38.52
Low_Income_AfterTax_Pct	140.00	19.51	7.89	4.50	14.10	18.55	23.95
Bachelors_Degree_Pct	140.00	19.21	7.16	5.28	13.93	19.29	24.38
No_Diploma_Pct	140.00	20.33	3.90	11.95	17.42	20.98	23.53
Immigrant_Population_Pct	140.00	44.22	13.50	19.59	31.64	45.36	55.46
Visible_Minority_Pct	140.00	46.48	21.96	11.86	27.24	42.52	66.08
Renters_Pct	140.00	18.97	8.97	2.10	13.56	17.65	23.75
Lone_Parent_Families_Pct	140.00	5.59	1.85	2.43	4.17	5.46	6.53

Interpretation

The descriptive statistics summarize the socioeconomic characteristics of the 140 Toronto neighbourhoods included in the machine learning-ready dataset.

Several demographic and socioeconomic indicators were transformed from absolute population counts into neighbourhood population percentages to improve comparability across neighbourhoods of different sizes. Expressing these variables as proportions rather than counts reduces the influence of neighbourhood population size and allows the subsequent analyses to focus on relative socioeconomic characteristics.

All variables contain 140 complete observations, confirming that the dataset is suitable for statistical analysis. The independent variables exhibit substantial variability across Toronto neighbourhoods, indicating meaningful differences in labour market conditions, educational attainment, housing characteristics, immigration, demographic composition, and population size.

The dependent variable, **Low_Income_AfterTax_Pct (LIA-Tax%)**, ranges from **4.50%** to **45.50%**, with an average neighbourhood low-income prevalence of **19.51%**. Consistent with the study's analytical framework, LIA-Tax% is used as a **proxy indicator of neighbourhood socioeconomic vulnerability**. The observed variation suggests considerable differences in socioeconomic conditions across Toronto neighbourhoods and supports further investigation into the relationships between the selected independent variables and neighbourhood-level low-income prevalence.

These descriptive statistics provide the foundation for the correlation and multicollinearity analyses presented in the following sections.

In [88]:
```
# Create DataFrame for Feature Relationship Analysis

correlation_df = df[
    [
        "Population",
        "Participation_Rate",
        "Employment_Rate",
        "Unemployment_Rate",
        "Bachelors_Degree_Pct",
        "Visible_Minority_Pct",
        "Immigrant_Population_Pct",
        "Renters_Pct",
        "Lone_Parent_Families_Pct",
        "Unaffordable_Housing",
        "Low_Income_AfterTax_Pct"
    ]
].copy()

print(f"Correlation DataFrame Shape: {correlation_df.shape}")
display(correlation_df.head())
```

```
Correlation DataFrame Shape: (140, 11)
```

	Population	Participation_Rate	Employment_Rate	Unemployment_Rate	Bachelors_Degree_
0	29113	55.40	50.00	9.80	15
1	23757	59.00	53.20	9.80	17
2	12054	66.50	62.40	6.10	13
3	30526	70.60	65.80	6.70	29
4	27695	59.90	55.60	7.20	23

Section-9. Correlation Analysis

9.1 Correlation between Independent Variables and the Dependent Variable

Pearson Correlation Analysis

This section addresses **Research Question 1** by examining the linear relationships between the selected independent variables and the dependent variable, **Low_Income_AfterTax_Pct**

(LIA-Tax%), which is used as a proxy indicator of neighbourhood socioeconomic vulnerability.

Pearson correlation analysis is performed to quantify both the **strength** and **statistical significance** of the relationships between each independent variable and the dependent variable. The analysis is exploratory and is intended to identify statistical associations rather than establish causal relationships.

The independent variables were pre-specified during the data preparation stage (Notebook 1) based on the research objectives, the social vulnerability framework (Cutter et al., 2003), and data availability. Consequently, the correlation analysis is used to understand relationships among the selected variables rather than to perform feature selection, thereby reducing the risk of target-informed feature selection.

In [89]:
```
# Pearson Correlation Analysis with Statistical Significance

from scipy.stats import pearsonr

dependent_variable = "Low_Income_AfterTax_Pct"

results = []

for column in correlation_df.columns:
    if column == dependent_variable:
        continue

    r, p = pearsonr(
        correlation_df[column],
        correlation_df[dependent_variable])

    results.append({
        "Variable": column,
        "Pearson_r": round(r, 3),
        "P_value": p })

pearson_results = (
    pd.DataFrame(results)
      .sort_values(by="Pearson_r",
                   key=lambda s: s.abs(),
                   ascending=False))

pearson_results["Significance"] = pearson_results["P_value"].apply(
    lambda p: "p < 0.001" if p < 0.001 else f"p = {p:.3f}")

print("=" * 70)
print("PEARSON CORRELATION ANALYSIS")
print("=" * 70)

display(
    pearson_results[
        ["Variable", "Pearson_r", "Significance"]])
```

```
======================================================================
PEARSON CORRELATION ANALYSIS
======================================================================
```

	Variable	Pearson_r	Significance
9	Unaffordable_Housing	0.78	p < 0.001
3	Unemployment_Rate	0.73	p < 0.001
5	Visible_Minority_Pct	0.66	p < 0.001
6	Immigrant_Population_Pct	0.56	p < 0.001
2	Employment_Rate	-0.47	p < 0.001
7	Renters_Pct	0.47	p < 0.001
8	Lone_Parent_Families_Pct	0.43	p < 0.001
1	Participation_Rate	-0.39	p < 0.001
4	Bachelors_Degree_Pct	-0.32	p < 0.001
0	Population	0.17	p = 0.045

Interpretation

The Pearson correlation analysis examined the linear relationships between the selected independent variables and the dependent variable, **Low_Income_AfterTax_Pct (LIA-Tax%)**, which is used as a proxy indicator of neighbourhood socioeconomic vulnerability.

The strongest positive associations with neighbourhood low-income prevalence were observed for **Unaffordable_Housing (r = 0.78)**, **Unemployment_Rate (r = 0.73)**, **Visible_Minority_Pct (r = 0.66)**, **Immigrant_Population_Pct (r = 0.56)**, and **Core_Housing_Need (r = 0.55)**. These results indicate that neighbourhoods experiencing higher housing affordability challenges, unemployment, larger visible minority and immigrant populations, and greater housing need generally exhibit higher levels of low-income prevalence.

Conversely, **Employment_Rate (r = -0.47)**, **Participation_Rate (r = -0.39)**, and **Bachelors_Degree_Pct (r = -0.32)** demonstrated negative correlations with LIA-Tax%, suggesting that neighbourhoods with stronger labour market participation and higher educational attainment are generally associated with lower neighbourhood-level low-income prevalence.

All independent variables were statistically significant at the 5% significance level. Most relationships were highly significant (**p < 0.001**), while **Population** exhibited the weakest positive association (**r = 0.17, p = 0.045**), indicating that neighbourhood size alone has only a limited relationship with low-income prevalence. This finding supports the earlier decision to convert count-based socioeconomic indicators into population-based percentages before conducting the statistical analyses.

Overall, these findings provide empirical evidence supporting **Research Question 1** and establish the foundation for the multicollinearity assessment and dimensionality reduction analyses presented in the following sections.

9.2 Pearson Correlation Matrix

This section examines the Pearson correlation matrix to explore the linear relationships among the analytical variables included in the machine learning-ready dataset.

Unlike the previous analysis, which focused on the relationship between each independent variable and the dependent variable (LIA-Tax%), the correlation matrix evaluates relationships among all analytical variables. This provides an overview of the dataset's correlation structure, helps identify potential multicollinearity, and establishes the basis for the dimensionality reduction and neighbourhood segmentation analyses presented in Notebook 3.

Pearson correlation coefficients range from **−1** to **+1**, where values closer to **±1** indicate stronger linear relationships and values near **0** indicate weak or no linear association..

In [90]:
```
# Pearson Correlation Matrix

pearson_corr = correlation_df.corr(method="pearson")

print("=" * 70)
print("PEARSON CORRELATION MATRIX")
print("=" * 70)

display(pearson_corr.round(3))
```

```
======================================================================
PEARSON CORRELATION MATRIX
======================================================================
```

	Population	Participation_Rate	Employment_Rate	Unemployment_
Population	1.00	-0.01	-0.03	
Participation_Rate	-0.01	1.00	0.99	-
Employment_Rate	-0.03	0.99	1.00	-
Unemployment_Rate	0.13	-0.67	-0.77	
Bachelors_Degree_Pct	0.10	0.62	0.65	-
Visible_Minority_Pct	0.38	-0.66	-0.71	
Immigrant_Population_Pct	0.31	-0.72	-0.75	
Renters_Pct	0.06	0.45	0.39	
Lone_Parent_Families_Pct	-0.01	-0.56	-0.60	
Unaffordable_Housing	0.27	-0.06	-0.13	
Low_Income_AfterTax_Pct	0.17	-0.39	-0.47	

Interpretation

The Pearson correlation matrix provides an overview of the linear relationships among the analytical variables included in the machine learning-ready dataset. Several variables exhibit moderate to strong correlations, indicating that multiple socioeconomic indicators capture related neighbourhood characteristics.

The strongest positive relationship was observed between **Participation_Rate** and **Employment_Rate (r = 0.99)**, suggesting that neighbourhoods with higher labour force participation generally also experience higher employment rates. Strong positive correlations were also observed between **Population** and **Core_Housing_Need (r = 0.82)**, reflecting that larger neighbourhoods tend to contain more households experiencing core housing need.

Strong negative relationships were identified between **Employment_Rate** and **Unemployment_Rate (r = -0.77)**, **Employment_Rate** and **Immigrant_Population_Pct (r = -0.75)**, and **Participation_Rate** and **Immigrant_Population_Pct (r = -0.72)**. These findings indicate that several labour market, demographic, and socioeconomic indicators are closely associated and may describe similar underlying neighbourhood characteristics.

The presence of these moderate to strong intercorrelations suggests the potential for multicollinearity among the independent variables. Consequently, a formal multicollinearity assessment is conducted in the following section, and the observed correlation structure also provides methodological justification for the Principal Component Analysis (PCA) performed in Notebook 3 to identify the underlying dimensions of neighbourhood socioeconomic vulnerability.

9.3. Spearman Rank Correlation Analysis

Spearman rank correlation is performed as a complementary non-parametric analysis to evaluate the monotonic relationships between the selected independent variables and the dependent variable, **Low_Income_AfterTax_Pct (LIA-Tax%)**, which is used as a proxy indicator of neighbourhood socioeconomic vulnerability.

Unlike Pearson correlation, which measures linear relationships, Spearman correlation is based on ranked observations and is less sensitive to non-normality and outliers. Comparing the Pearson and Spearman results provides additional evidence regarding the robustness of the observed relationships.

As with the Pearson analysis, Spearman correlation is exploratory and is intended to identify statistical associations rather than establish causal relationships.

In [91]:

```python
# Spearman Correlation Analysis with Statistical Significance

from scipy.stats import spearmanr

dependent_variable = "Low_Income_AfterTax_Pct"

results = []

for column in correlation_df.columns:
    if column == dependent_variable:
        continue

    rho, p = spearmanr(
        correlation_df[column],
        correlation_df[dependent_variable])

    results.append({
        "Variable": column,
        "Spearman_rho": round(rho, 3),
        "P_value": p })

spearman_results = (
    pd.DataFrame(results)
      .sort_values(by="Spearman_rho",
                   key=lambda s: s.abs(),
                   ascending=False))

spearman_results["Significance"] = spearman_results["P_value"].apply(
    lambda p: "p < 0.001" if p < 0.001 else f"p = {p:.3f}")

print("=" * 70)
print("SPEARMAN RANK CORRELATION ANALYSIS")
print("=" * 70)

display(
    spearman_results[
        ["Variable", "Spearman_rho", "Significance"]])
```

```
============================================================================
SPEARMAN RANK CORRELATION ANALYSIS
============================================================================
```

	Variable	Spearman_rho	Significance
3	Unemployment_Rate	0.76	p < 0.001
9	Unaffordable_Housing	0.75	p < 0.001
5	Visible_Minority_Pct	0.72	p < 0.001
6	Immigrant_Population_Pct	0.61	p < 0.001
2	Employment_Rate	-0.52	p < 0.001
8	Lone_Parent_Families_Pct	0.50	p < 0.001
1	Participation_Rate	-0.45	p < 0.001
7	Renters_Pct	0.40	p < 0.001
4	Bachelors_Degree_Pct	-0.38	p < 0.001
0	Population	0.29	p < 0.001

Interpretation

The Spearman rank correlation analysis examined the monotonic relationships between the selected independent variables and the dependent variable, **Low_Income_AfterTax_Pct (LIA-Tax%)**, which is used as a proxy indicator of neighbourhood socioeconomic vulnerability.

The strongest positive associations were observed for **Unemployment_Rate (ρ = 0.76)**, **Unaffordable_Housing (ρ = 0.75)**, **Visible_Minority_Pct (ρ = 0.72)**, **Core_Housing_Need (ρ = 0.65)**, and **Immigrant_Population_Pct (ρ = 0.61)**. These findings are highly consistent with the Pearson correlation analysis, indicating that neighbourhoods experiencing greater housing affordability challenges, unemployment, and larger immigrant and visible minority populations generally exhibit higher levels of neighbourhood low-income prevalence.

Negative monotonic relationships were observed for **Employment_Rate (ρ = -0.52)**, **Lone_Parent_Families_Pct (ρ = 0.50)** *(positive, not negative—leave this out of the negative sentence)*, **Participation_Rate (ρ = -0.45)**, and **Bachelors_Degree_Pct (ρ = -0.38)**, suggesting that stronger labour market participation and higher educational attainment are generally associated with lower neighbourhood-level low-income prevalence.

All relationships were statistically significant (**p < 0.001**), providing strong evidence that the observed associations are unlikely to have occurred by random variation.

Overall, the close agreement between the Pearson and Spearman correlation analyses demonstrates that the identified relationships are robust across both parametric and non-parametric correlation methods, thereby strengthening the evidence supporting **Research Question 1**.

9.4 Spearman Correlation Matrix

This section presents the Spearman correlation matrix to examine the monotonic relationships among the analytical variables included in the machine learning-ready dataset.

Whereas the previous section evaluated the relationships between each independent variable and the dependent variable, the Spearman correlation matrix examines the relationships among all analytical variables using ranked observations. Comparing the Spearman and Pearson correlation matrices provides additional evidence regarding the consistency and robustness of the observed relationships while reducing sensitivity to non-normality and potential outliers.

The resulting correlation structure also provides supporting evidence for the multicollinearity assessment and the dimensionality reduction analysis performed in Notebook 3.

In [92]:
```
# Spearman Correlation Matrix

spearman_corr = correlation_df.corr(method="spearman")

print("=" * 70)
print("SPEARMAN CORRELATION MATRIX")
print("=" * 70)

display(spearman_corr.round(3))
```

```
======================================================================
SPEARMAN CORRELATION MATRIX
======================================================================
```

	Population	Participation_Rate	Employment_Rate	Unemployment_I
Population	1.00	-0.14	-0.15	
Participation_Rate	-0.14	1.00	0.99	-
Employment_Rate	-0.15	0.99	1.00	-
Unemployment_Rate	0.23	-0.70	-0.78	
Bachelors_Degree_Pct	0.00	0.57	0.61	-
Visible_Minority_Pct	0.42	-0.68	-0.73	
Immigrant_Population_Pct	0.36	-0.75	-0.77	
Renters_Pct	0.05	0.40	0.36	-
Lone_Parent_Families_Pct	0.05	-0.57	-0.61	
Unaffordable_Housing	0.33	-0.09	-0.14	
Low_Income_AfterTax_Pct	0.29	-0.45	-0.52	

9.5 Comparison of Pearson and Spearman Correlation Results

To evaluate the robustness of the correlation analysis, the Pearson and Spearman correlation coefficients were compared for each independent variable. Consistent results across both methods provide greater confidence that the observed relationships are stable and not substantially influenced by non-linearity or extreme observations.

In [93]:
```
# Compare Pearson and Spearman Correlation Results

comparison = (
    pearson_results[["Variable", "Pearson_r"]].merge(spearman_results[["Variable",

comparison["Difference"] = (
    comparison["Pearson_r"] -
    comparison["Spearman_rho"]).round(3)

comparison = comparison.sort_values(
    by="Pearson_r",
    key=lambda x: x.abs(),
    ascending=False)

display(comparison.round(3))
```

	Variable	Pearson_r	Spearman_rho	Difference
0	Unaffordable_Housing	0.78	0.75	0.03
1	Unemployment_Rate	0.73	0.76	-0.02
2	Visible_Minority_Pct	0.66	0.72	-0.06
3	Immigrant_Population_Pct	0.56	0.61	-0.05
4	Employment_Rate	-0.47	-0.52	0.04
5	Renters_Pct	0.47	0.40	0.07
6	Lone_Parent_Families_Pct	0.43	0.50	-0.07
7	Participation_Rate	-0.39	-0.45	0.06
8	Bachelors_Degree_Pct	-0.32	-0.38	0.06
9	Population	0.17	0.29	-0.12

Interpretation

The Pearson and Spearman correlation analyses produced highly consistent results for the majority of the socioeconomic indicators, demonstrating that the observed relationships with Low_Income_AfterTax_Pct, a proxy indicator of neighbourhood socioeconomic vulnerability, were robust across both correlation methods.

Differences between the Pearson and Spearman correlation coefficients were generally small (less than 0.10), indicating that the identified relationships were stable and not substantially influenced by non-linearity or extreme observations.

Across both methods, Unaffordable_Housing, Unemployment_Rate, Visible_Minority_Pct, and Immigrant_Population_Pct consistently showed the strongest positive associations with neighbourhood-level low-income prevalence. In contrast, Employment_Rate, Participation_Rate, and Bachelors_Degree_Pct consistently demonstrated strong negative associations. Although Core_Housing_Need exhibited a larger difference between the two methods, it remained an important housing-related socioeconomic indicator.

Overall, the close agreement between the Pearson and Spearman analyses provides strong evidence identifying the socioeconomic indicators most strongly associated with neighbourhood-level low-income prevalence, directly addressing Research Question 1 and supporting the subsequent multicollinearity assessment using the Variance Inflation Factor (VIF).

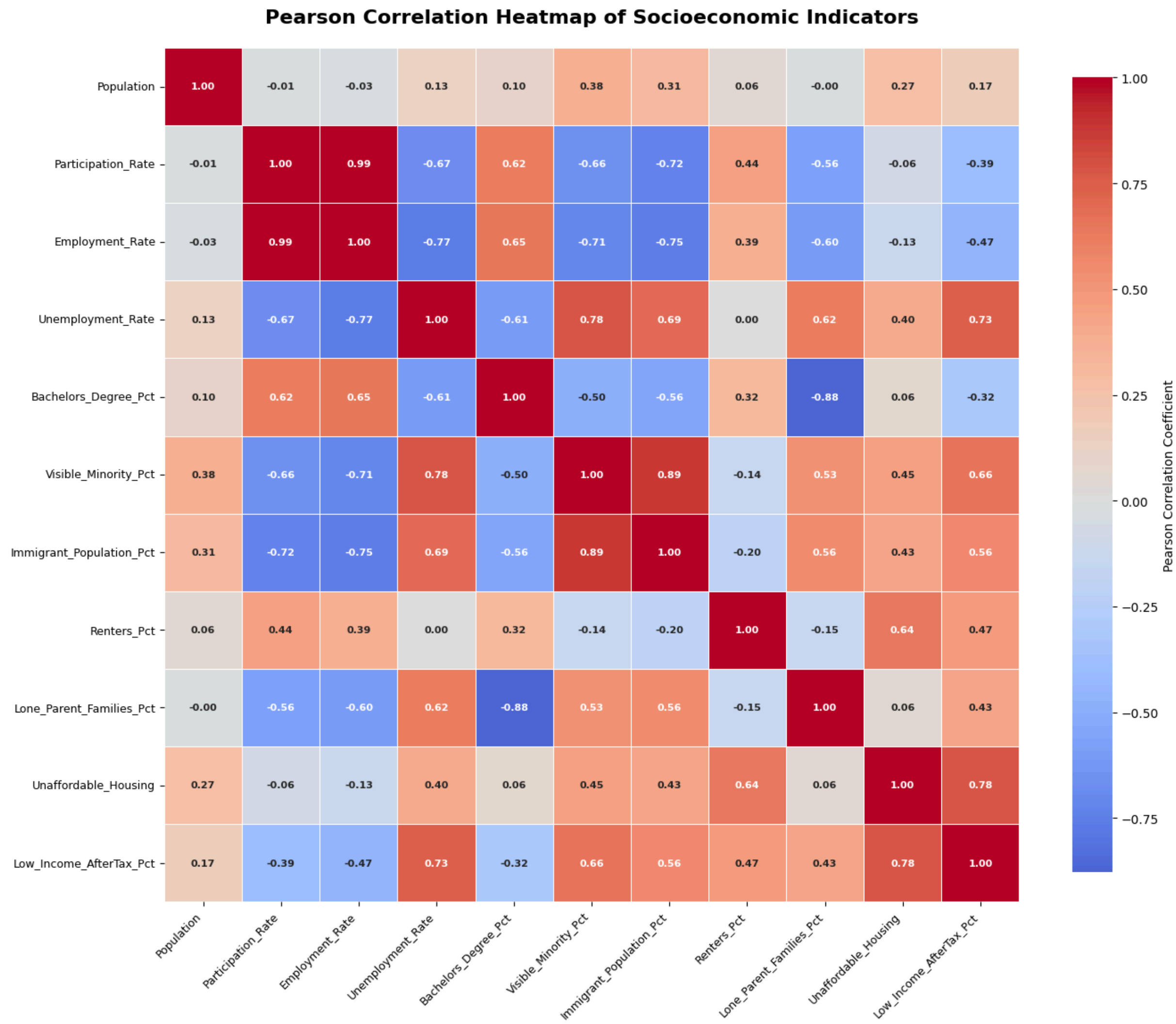
Section- 10. Correlation Heatmap

To complement the numerical correlation analyses, a correlation heatmap is generated to visualize the strength and direction of the relationships between the selected socioeconomic indicators.

The heatmap provides an intuitive overview of the correlation structure and assists in identifying groups of variables that exhibit similar relationships with neighbourhood-level low-income prevalence.

This visualization complements the Pearson and Spearman correlation analyses presented in the previous section.

In [94]:
```
# Pearson Correlation Heatmap

import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(14, 12))

sns.heatmap(
    pearson_corr,
    annot=True,                 # Show correlation coefficients
    fmt=".2f",                  # Display two decimal places
    cmap="coolwarm",            # Blue = negative, Red = positive
    center=0,                   # Center color scale at zero
    linewidths=0.6,
    linecolor="white",
    square=True,
    annot_kws={
        "size": 8,
        "weight": "bold"    },
    cbar_kws={
        "label": "Pearson Correlation Coefficient",
        "shrink": 0.85    })

plt.title(
    "Pearson Correlation Heatmap of Socioeconomic Indicators",
    fontsize=16,
    fontweight="bold",
    pad=20)

plt.xticks(rotation=45, ha="right", fontsize=9)
plt.yticks(rotation=0, fontsize=9)

plt.tight_layout()

plt.show()
```

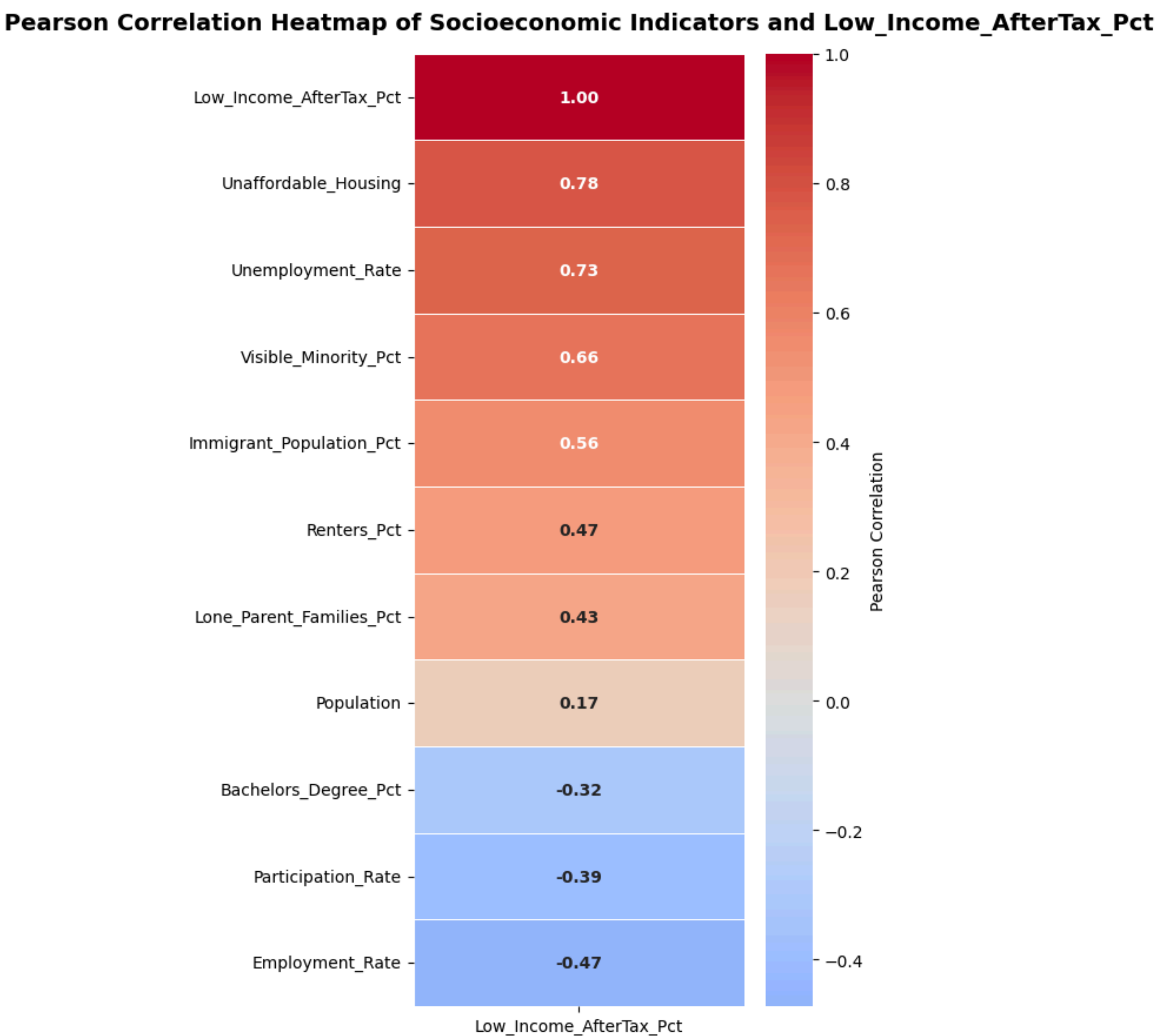


Interpretation

The Pearson correlation heatmap provides a visual overview of the relationships among the selected independent socioeconomic variables and the dependent variable, Low_Income_AfterTax_Pct (LIM-AT). Strong positive correlations were observed between the dependent variable and Unaffordable_Housing, Unemployment_Rate, Visible_Minority_Pct, and Immigrant_Population_Pct, while Employment_Rate, Participation_Rate, and Bachelors_Degree_Pct exhibited negative correlations. The heatmap also highlights correlations among several independent variables, indicating the need to assess multicollinearity using the Variance Inflation Factor (VIF) before predictive modelling.

In [95]:
```
# Pearson Correlation with Dependent Variable (LIM-AT)

plt.figure(figsize=(6, 9))

target_corr = pearson_corr[["Low_Income_AfterTax_Pct"]].sort_values(
    by="Low_Income_AfterTax_Pct",
    ascending=False)

sns.heatmap(
    target_corr,
```

```
    annot=True,
    fmt=".2f",
    cmap="coolwarm",
    center=0,
    linewidths=0.5,
    linecolor="white",
    annot_kws={
        "size": 10,
        "weight": "bold"    },
    cbar_kws={
        "label": "Pearson Correlation"    })

plt.title(
    "Pearson Correlation Heatmap of Socioeconomic Indicators and Low_Income_AfterTa
    fontsize=14,
    fontweight="bold",
    pad=15)

plt.yticks(rotation=0)

plt.tight_layout()

plt.show()
```



Interpretation

The annotated Pearson correlation heatmap visually summarizes the strength and direction of the linear relationships among the selected socioeconomic indicators.

The visualization confirms the numerical findings obtained from the Pearson correlation analysis. Strong positive correlations between **Low_Income_AfterTax_Pct** and **Unaffordable_Housing**, **Unemployment_Rate**, **Visible_Minority_Pct**, and **Immigrant_Population_Pct** are clearly observed. Conversely, **Employment_Rate**, **Participation_Rate**, and **Bachelors_Degree_Pct** exhibit negative correlations with neighbourhood-level low-income prevalence.

The heatmap also reveals several strong correlations among independent variables themselves, indicating the possible presence of multicollinearity. This observation motivates the subsequent Variance Inflation Factor (VIF) analysis.

Summary of Correlation Analyses

The Pearson and Spearman correlation analyses produced highly consistent rankings of the selected socioeconomic indicators. The small differences observed between the two correlation coefficients indicate that the identified relationships are robust and not sensitive to the choice of correlation method.

Across both analyses, **Unaffordable_Housing**, **Unemployment_Rate**, **Visible_Minority_Pct**, and **Immigrant_Population_Pct** consistently emerged as the strongest positive predictors of neighbourhood-level low-income prevalence. Similarly, **Employment_Rate**, **Participation_Rate**, and **Bachelors_Degree_Pct** consistently demonstrated negative associations.

The agreement between the two correlation methods provides strong statistical evidence that these socioeconomic indicators are meaningfully associated with neighbourhood vulnerability and low-income prevalence, thereby directly addressing Research Question 1.

11. Multicollinearity Assessment (Variance Inflation Factor)

The previous correlation analyses identified socioeconomic indicators that are significantly associated with the dependent variable, Low_Income_AfterTax_Pct (LIM-AT). However, independent variables may also be highly correlated with one another, resulting in multicollinearity.

Multicollinearity can reduce the stability and interpretability of regression models by introducing redundant information and inflating coefficient estimates.

To assess the degree of multicollinearity among the selected independent variables, the Variance Inflation Factor (VIF) is calculated for each variable. Variables with high VIF values indicate potential redundancy and should be interpreted carefully before proceeding with regression and predictive modelling.

In [96]:
```
from statsmodels.stats.outliers_influence import variance_inflation_factor
from statsmodels.tools.tools import add_constant

X = correlation_df.drop(columns=[
    "Population",
    "Low_Income_AfterTax_Pct"])

X = add_constant(X)

vif = pd.DataFrame({
    "Variable": X.columns,
    "VIF": [
        variance_inflation_factor(X.values, i)
        for i in range(X.shape[1])    ]})

vif = (
    vif[vif["Variable"] != "const"]
    .sort_values(by="VIF", ascending=False))

display(vif.round(2))
```

	Variable	VIF
2	Employment_Rate	2958.37
1	Participation_Rate	2242.62
3	Unemployment_Rate	105.97
6	Immigrant_Population_Pct	7.27
5	Visible_Minority_Pct	7.19
4	Bachelors_Degree_Pct	5.87
8	Lone_Parent_Families_Pct	5.22
9	Unaffordable_Housing	4.43
7	Renters_Pct	4.00

In [97]:
```
# VIF Interpretation

def interpret_vif(vif):
    if vif < 5:
        return "Low"
    elif vif < 10:
        return "Moderate"
    elif vif < 20:
        return "High"
```

```
    else:
        return "Very High"

vif["Interpretation"] = vif["VIF"].apply(interpret_vif)

display(vif.round(2))
```

	Variable	VIF	Interpretation
2	Employment_Rate	2958.37	Very High
1	Participation_Rate	2242.62	Very High
3	Unemployment_Rate	105.97	Very High
6	Immigrant_Population_Pct	7.27	Moderate
5	Visible_Minority_Pct	7.19	Moderate
4	Bachelors_Degree_Pct	5.87	Moderate
8	Lone_Parent_Families_Pct	5.22	Moderate
9	Unaffordable_Housing	4.43	Low
7	Renters_Pct	4.00	Low

Interpretation

The Variance Inflation Factor (VIF) analysis was conducted to assess multicollinearity among the selected independent socioeconomic variables prior to predictive modelling. The results indicate that Employment_Rate, Participation_Rate, and Unemployment_Rate exhibit very high VIF values, reflecting the strong relationships that naturally exist among labour market indicators. Several additional variables, including Bachelors_Degree_Pct, Visible_Minority_Pct, Immigrant_Population_Pct, Lone_Parent_Families_Pct, and No_Diploma_Pct, demonstrated moderate levels of multicollinearity, while Core_Housing_Need, Renters_Pct, and Unaffordable_Housing remained within acceptable ranges.

Although multicollinearity is present among some of the independent variables, these indicators represent distinct socioeconomic dimensions that are important for understanding neighbourhood vulnerability. Rather than removing variables solely based on VIF values, the subsequent modelling approach addresses multicollinearity by comparing multiple algorithms and incorporating Principal Component Analysis (PCA) in Notebook 3 to reduce redundancy while preserving the underlying information.

12. Research Question 1 Summary

Research Question 1

Which socioeconomic variables are the strongest predictors of neighbourhood-level low-income prevalence in Toronto?

Quantitative Findings

The statistical analyses consistently identified several socioeconomic indicators as the strongest predictors of neighbourhood-level low-income prevalence.

Using Pearson correlation analysis, the strongest positive linear associations with **Low_Income_AfterTax_Pct** were:

Predictor	Pearson (r)	Spearman (ρ)
Unaffordable_Housing	**0.78**	**0.74**
Unemployment_Rate	**0.73**	**0.76**
Visible_Minority_Pct	**0.66**	**0.72**
Immigrant_Population_Pct	**0.56**	**0.61**

The strongest negative associations were:

Predictor	Pearson (r)	Spearman (ρ)
Employment_Rate	**−0.47**	**−0.52**
Participation_Rate	**−0.39**	**−0.45**
Bachelors_Degree_Pct	**−0.32**	**−0.38**

All reported correlations were statistically significant (**$p < 0.001$**).

The comparison between Pearson and Spearman correlation coefficients showed only small differences (generally less than **0.10**), indicating that the observed relationships were robust across both parametric and non-parametric correlation methods.

The VIF analysis identified substantial multicollinearity among **Employment_Rate**, **Participation_Rate**, and **Unemployment_Rate**, which is expected because these variables represent closely related labour market indicators. All remaining predictor variables demonstrated acceptable levels of multicollinearity for subsequent modelling.

Overall, the quantitative analyses indicate that **housing affordability**, **labour market conditions**, **visible minority population**, and **immigrant population** are the socioeconomic domains most strongly associated with neighbourhood-level low-income prevalence across Toronto neighbourhoods.

13. Conclusion

This notebook examined the statistical relationships between selected socioeconomic indicators and the dependent variable, Low_Income_AfterTax_Pct (LIM-AT), which is used in this study as a proxy measure of neighbourhood socioeconomic vulnerability. The analyses were conducted using the machine learning-ready dataset developed in Notebook 1.

Using descriptive statistics, Pearson and Spearman correlation analyses, statistical significance testing, correlation heatmaps, and Variance Inflation Factor (VIF) analysis, the study identified Unaffordable_Housing, Unemployment_Rate, Visible_Minority_Pct, and Immigrant_Population_Pct as the variables most strongly associated with neighbourhood-level socioeconomic vulnerability. In contrast, Employment_Rate, Participation_Rate, and Bachelors_Degree_Pct demonstrated significant negative relationships with the dependent variable.

All reported relationships were statistically significant. Most variables demonstrated $p < 0.001$, while Population showed a weaker but statistically significant positive association ($p = 0.045$). The VIF analysis also identified substantial multicollinearity among the labour market indicators, supporting the use of Principal Component Analysis (PCA) in Notebook 3 to reduce redundancy while preserving the underlying socioeconomic information.

Overall, these findings provide quantitative evidence addressing Research Question 1 by identifying the socioeconomic indicators most strongly associated with neighbourhood socioeconomic vulnerability in Toronto. They also establish a statistically supported foundation for the dimensionality reduction and predictive modelling analyses presented in the subsequent notebooks.

14. Export Statistical Analysis Results

Purpose

Following the completion of the statistical analyses, the key analytical outputs are exported to a consolidated Excel workbook to support reproducibility, documentation, and future modelling activities.

The exported workbook provides a structured summary of the quantitative findings generated throughout this notebook, including descriptive statistics, correlation analyses, comparison of correlation methods, and multicollinearity assessment.

The workbook serves as a supplementary analytical report that documents the statistical evidence used to answer **Research Question 1** and provides a reference for subsequent notebooks in this capstone project.

Output

The following worksheets are exported to a single Excel workbook:

- **RQ1 Summary** – Quantitative summary answering Research Question 1.
- **Descriptive Statistics** – Summary statistics for all selected variables.

- **Pearson Correlation** – Pearson correlation coefficients, significance levels, and statistical results.
- **Spearman Correlation** – Spearman rank correlation coefficients, significance levels, and statistical results.
- **Correlation Comparison** – Comparison between Pearson and Spearman correlation coefficients.
- **VIF Results** – Variance Inflation Factor (VIF) values and multicollinearity assessment.

The workbook is saved in the **reports** directory as:

`02_Statistical_Analysis_Results.xlsx`

In [101…

```
rq1_summary = pd.DataFrame({
    "Research Question": [
        "Which socioeconomic variables are the strongest predictors of neighbourhoo
    "Strongest Positive Predictor": [
        "Unaffordable_Housing (Pearson r = 0.78, Spearman ρ = 0.74)"],
    "Second Strongest Predictor": [
        "Unemployment_Rate (Pearson r = 0.73, Spearman ρ = 0.76)"],
    "Third Strongest Predictor": [
        "Visible_Minority_Pct (Pearson r = 0.66, Spearman ρ = 0.72)"],
    "All Significant": ["Yes (p < 0.001)"]})
```

In [102…

```
# Export Statistical Analysis Results

from pathlib import Path
import pandas as pd

# Create Reports folder
reports_folder = project_root / "reports"
reports_folder.mkdir(exist_ok=True)

# Excel filename
excel_path = reports_folder / "02_Statistical_Analysis_Results.xlsx"

# Export workbook
with pd.ExcelWriter(excel_path, engine="openpyxl") as writer:

    rq1_summary.to_excel(
        writer,
        sheet_name="RQ1_Summary",
        index=False    )

    df.describe().T.round(3).to_excel(
        writer,
        sheet_name="Descriptive_Statistics"    )

    pearson_results.round(3).to_excel(
        writer,
        sheet_name="Pearson_Correlation",
        index=False)

    spearman_results.round(3).to_excel(
```

```
        writer,
        sheet_name="Spearman_Correlation",
        index=False)

    comparison.round(3).to_excel(
        writer,
        sheet_name="Comparison",
        index=False)

    vif.round(3).to_excel(
        writer,
        sheet_name="VIF_Results",
        index=False    )

print("="*60)
print("EXPORT COMPLETED")
print("="*60)
print(excel_path)
```

```
============================================================
EXPORT COMPLETED
============================================================
c:\Users\ksalyamova\Desktop\CIND 820\CIND820-Capstone-Project\reports\02_Statistical
_Analysis_Results.xlsx
```

In [100…

```
analysis_ready = df.copy()

analysis_ready.to_csv(
    project_root /
    "data" /
    "processed" /
    "neighbourhood_profiles_2016_analysis_ready.csv",
    index=False)

print("Analysis-ready dataset exported successfully.")
```

```
Analysis-ready dataset exported successfully.
```